

Lecture 1: Real Numbers

AMA3707 Real Analysis

1 Real Numbers

1.1 Numbers and Fields

Definition 1.1.1 (Natural Numbers). The *natural numbers* are the positive integers:

$$\mathbb{N} = \{1, 2, 3, \dots\}.$$

In these notes, $0 \notin \mathbb{N}$.

Definition 1.1.2 (Integers). The *integers* are the whole numbers, including the positive integers, the negative integers, and zero:

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}.$$

Definition 1.1.3 (Rational Numbers). A real number is called *rational* if it can be expressed as a quotient of two integers. The set of rational numbers is

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}.$$